Question 1

At time t, a country inherits from the previous period a debt-to-GDP ratio of b_{t-1}^{0}. The economy has a primary surplus and the growth of rate is higher than the real interest rate. Which of the two figures below represents the debt-to-GDP line? Will this economy converge to the steady state without any intervention? Explain. If the government decides, instead, to stabilize the debt-to-GDP ratio at the level b_{t-1}^{0} in t, what type of policy could they use? Name two of them.

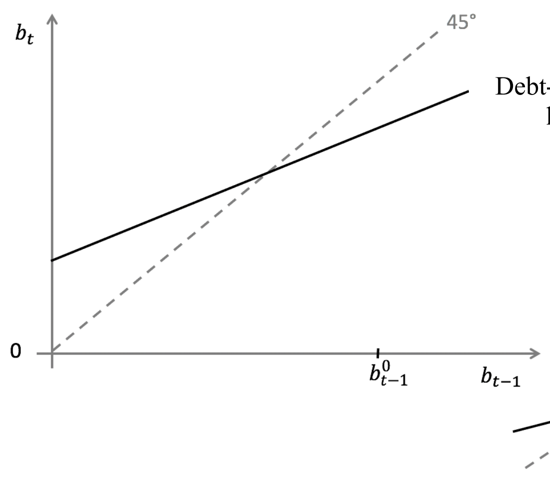

Figure 1

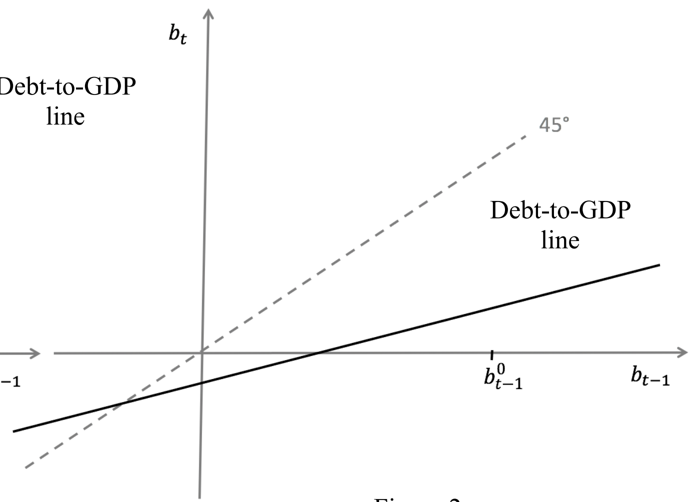

Figure 2

Figure 2. The dynamics of the debt-to-GDP ratio are given by the equation of the debt-to-GDP line: $b_t=(1+r-g)*b_{t-1}+d$, where b=B/Y, and d=(G-T)/Y is the primary deficit to GDP. It is straightforward to verify that the vertical intercept of the debt-to-GDP line is equal to the value of d. In the economy under consideration, the government runs in a primary surplus, i.e. tax revenues are greater than public expenditure, G<T and d<0. The intercept d will therefore be a negative value as in figure 2, and opposed to figure 1.

Furthermore, since the debt-to-GDP line is flatter than the one at 45 degree passing through the origin, we know that r<g. Absent any interventions, the economy will converge to a steady state which corresponds to the point of intersection between the debt-to-GDP line and the 45 degree line passing through the origin. In Figure 2, the steady state is a negative value. Since r<g, the debt-to-GDP ratio tends to decrease over time. This diminishing trend of the debt-to-GDP ratio is reinforced by the fact that d<0. The economy will have a stationary value of negative public debt, that is, it will save. The existence of a negative stationary value is due to the fact that the primary surplus is counterbalanced by an interest rate received on savings which is lower than the growth rate of Y and, therefore, the rate at which saving decreases in relation to the GDP.

If the government decides that it wants to stabilize the debt/GDP ratio at level b_{t-1}^0 from time t onwards, it can either: 1) increase the primary deficit to a level d>0 which causes the debt-to-GDP line and the 45 degree line to intersect at the value di $b = b_{t-1}^0$, or 2) increase r (leaving g unchanged) so that r>g and, thus, the slope of the debt-to-GDP line is such that it intersects with the 45 degree line at the value $b = b_{t-1}^0$.

Question 2

In Macrolandia, there are only one-year and two-year bonds. The current and the expected future inflation rates are both zero, so that the real and the nominal interest rates are equal. In period t+1 ("the future"), a standard IS-LM model describes the functioning of the economy. In particular, in the year t+1 consumption depends only on disposable income at t+1, and investment on the interest rate an output at t+1 only. Turning now to the current period ("the present"), consumption is increasing in in both current and expected future disposable income, $C_t = C(Y_t - \bar{T}_t, Y_{t+1}^e - \bar{T}_{t+1}^e)$, investment depends positively on current and expected future income levels, and negatively on the current and expected future levels of short-term interest rates, the demand for money has the usual functional form, and G and T are exogenous.

Starting from an initial equilibrium, at time t the government of Macrolandia announces a permanent reduction of its purchases of goods and services, $\Delta\bar{G}_t = \Delta\bar{G}_{t+1} < 0$.
The central bank conducts monetary policy choosing a value for the interest rate in each period and decides to intervene to keep the level of production fixed only in one of the periods, t or t+1, in which it deviates more from the its initial equilibrium value.

Describe the effects of these announcements, which are credible and take individuals by surprise, on the equilibrium value of income and the interest rate in period t and period t+1. Will the yield to maturity on two-year bonds at time t change, i_{2t}? Why?

The reduction in public spending shifts the IS to the left only for the period in which this decrease occurs. The IS line at t is steeper than at t+1 since changes have a smaller effect on Y when it is jointly determined by the changes in present but also expected future variables. Since the change in G is identical at t and t+1 but the multiplier is greater in period t+1, the central bank will intervene to keep the level of output fixed only in period t+1.

At t+1, therefore, the central bank conducts an expansionary monetary policy, reducing the policy rate, which leads the LM to shift downwards until it crosses the new IS at the point where the equilibrium Y value equals the initial equilibrium value. In equilibrium at t+1, income has not changed and the interest rate is lower.

At t, the reduction in public spending and expectations of a lower expected future interest rate have opposite effects on aggregate demand. It is not possible to determine the direction of the shift in the IS curve nor the resulting change in equilibrium income. As we have already argued, the central bank does not intervene during this period and, therefore, the interest rate at t remains unchanged.

The yield to maturity of two-year bonds is approximately the average of the present and expected future short-term rates. Since the future short rate falls while the current short rate stays the same, it will fall by half the change in the future short rate.

Question 3

Consider an open economy freely trading goods, services and financial assets with the rest of the world, in a fixed exchange rate regime. Domestic and foreign prices are constant and, for simplicity, both equal to 1 (P=P*=1). Initially, the country is in an equilibrium and trade is balanced. In this equilibrium, domestic and foreign interest rates are equal and market participants expect that the exchange rate will be kept fixed at the current level also in the future, $E^e = \bar{E}_o$.

Suppose that financial markets start fearing an impeding devaluation and that the expectations about the future value of the exchange rate goes from $E^e = \bar{E}_o$ to $E^e = \bar{E}_1$, where $\bar{E}_1 < \bar{E}_o$.

Analyze and explain the effect of this change of expectations on the equilibrium values of the income, interest rate and net exports, assuming that the central bank is determined to keep the exchange rate fixed at the initial equilibrium level $\bar{E}_o$.

The line representing the interest rate parity condition – in the graph with the domestic interest rate on the vertical axis and the exchange rate on the horizontal axis – shifts up and to the left. If the interest rate remains at its initial level, the exchange rate depreciates immediately. The expected devaluation of the national currency brings the expected return of foreign securities above that of domestic securities, all other variables being equal. If the central bank is determined to keep the exchange rate constant at the initial equilibrium level, it will have to raise the domestic interest rate above the foreign rate by an amount equal to the expected devaluation of the domestic currency.

Thus, the LM shifts upwards and crosses the IS line further to the left. In the new equilibrium, it follows that income will be lower and the interest rate higher. Initially, the country under consideration was in trade balance (NX=0). Since net exports increase, as domestic income is lower and the exchange rate has remained unchanged, net exports will be positive (trade surplus) in the new equilibrium.

Question 4

True or False?

Motivate your answer, in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

"In the economy studied before (in question 3), suppose that the exchange rate regime is flexible instead of fixed (whereas the rest of the assumptions do not change). In this case, it is not possible to establish the effects of a depreciation of the future expected exchange rate on the trade balance because the change in the equilibrium value of net exports is ambiguous."

FALSE. If the exchange rate regime in the economy under consideration is flexible, rather than fixed, expectations that the expected future exchange rate will decline will lead to a positive trade balance. The interest rate parity line shifts up and to the left, as before, but now the exchange rate depreciates immediately. This leads to an increase in net exports (due to the Marshall-Lerner condition) which causes the aggregate demand line (ZZ line) to shift up in the goods market; while the line DD (=C+I+G) remains unchanged since it does not depend on E. Starting from the point where NX=0, the slope of the DD line (flatter than the 45 degree line but steeper than the ZZ line) implies that the new line ZZ' crosses the 45 degree line to the left of the intersection of ZZ' and DD. Therefore, at the new equilibrium point, NX>0 and, thus, net exports improve.